

GED Science Practice Test 18

PART A — QUESTIONS

GED-SCI18-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether light exposure during the night affects sleep quality. Group A sleeps in a dark room. Group B sleeps in a room with a small lamp on all night. Both groups follow the same bedtime schedule for one week. Sleep quality is measured using the same score scale, where a higher score means better sleep quality.

Group	Night condition	Average sleep quality score
A	Dark room	82
B	Small lamp on	67

Which conclusion is best supported by the data?

A) Sleeping with a small lamp on was associated with lower sleep quality.

B) Sleeping with a small lamp on was associated with higher sleep quality.

C) Light exposure had no relationship to sleep quality.

D) Both groups had exactly the same sleep quality score.

GED-SCI18-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the gravitational potential energy of four objects. Gravitational potential energy can be calculated as $\text{weight} \times \text{height}$.

Object	Weight (N)	Height (m)
A	18	5
B	30	2
C	12	9
D	45	1

Which object has the greatest gravitational potential energy?

A) Object A

B) Object B

C) Object C

D) Object D

GED-SCI18-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A region has many abandoned coal mines. Rainwater can move through exposed rock and carry acidic runoff into nearby streams. One cleanup plan covers exposed rock, treats acidic water, and restores vegetation around streambanks.

Which outcome would best support the cleanup plan?

- A) Stream pH moves closer to neutral and vegetation increases near streambanks.
- B) Stream water becomes more acidic after treatment.
- C) Exposed rock produces unlimited coal.
- D) Vegetation prevents all minerals from existing.

GED-SCI18-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The hypothalamus helps regulate body temperature, hunger, thirst, and hormone signals. It receives information from the body and helps trigger responses that keep internal conditions within a normal range.

Which statement best describes the hypothalamus?

- A) It helps coordinate homeostatic responses.
- B) It filters urine inside the kidney.
- C) It produces lava at plate boundaries.
- D) It separates white light into colors.

GED-SCI18-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A box has a weight of 35 N and is lifted to a shelf 4 meters above the floor.

Gravitational potential energy = weight $\times$ height

What is the box's gravitational potential energy in joules?

Type your answer as a whole number.

GED-SCI18-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how wheel diameter affects the distance a model car travels after one full turn of the wheel. The same car body and same surface are used in every trial. Only wheel diameter changes.

Which TWO factors should be kept constant?

- A) Car body
- B) Surface used for testing
- C) Wheel diameter
- D) Distance traveled
- E) Final measured value

GED-SCI18-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some genetic conditions are caused by changes in the number or structure of chromosomes rather than by a single base change in one gene. A chart showing the chromosomes from a cell is called a [D1], and it can help identify [D2].

Dropdowns:

D1) karyotype | pulley | aquifer

D2) chromosome abnormalities | sound frequency | rock porosity

GED-SCI18-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student observes a shadow on a wall. When an object blocks a light source, light cannot pass through the object, so a darker region forms behind it.

Which property of light mainly explains the shadow?

- A) Light travels in straight paths unless it is reflected, absorbed, or bent.
- B) Light is made only by living cells.
- C) Light must pass through every solid object.
- D) Light stops having energy when it reaches air.

GED-SCI18-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A seismologist explains that P waves travel through solids and liquids, but S waves travel only through solids. When earthquake waves pass through Earth, S waves do not pass through the outer core.

Which conclusion is best supported by this evidence?

- A) Earth's outer core is liquid.
- B) Earth's outer core is made only of air.
- C) S waves travel faster through liquids than solids.
- D) Earth has no internal layers.

GED-SCI18-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor measures average blood lactate after two exercise sessions.

Exercise session | Average blood lactate (mmol/L)

Light cycling | 1.8

High-intensity cycling | 6.4

Which statement is best supported by the data?

- A) High-intensity cycling was associated with higher blood lactate.
- B) Light cycling was associated with higher blood lactate.
- C) Both exercise sessions produced the same lactate level.
- D) Lactate cannot be measured in blood.

GED-SCI18-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Length of the crank handle
- B) Force needed to turn the crank
- C) Same load attached to the crank system
- D) Color of the crank handle

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI18-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The inner ear contains structures that detect head movement and body position. These signals are sent to the brain, which uses them along with visual information and muscle feedback to help maintain balance.

Which function is mainly described?

- A) Sensing body position and helping maintain balance
- B) Breaking down fats with bile
- C) Filtering nitrogen waste from blood
- D) Forming sedimentary rock layers

GED-SCI18-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student uses a rubber band to launch a small object. Pulling the rubber band stores elastic potential energy. When the rubber band is released, much of that stored energy becomes kinetic energy of the object.

Which energy conversion is mainly described?

- A) Elastic potential energy changing into kinetic energy
- B) Chemical energy changing into genetic information
- C) Sound energy changing into rock layers
- D) Nuclear energy changing into groundwater

GED-SCI18-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A town reduces annual water use from 12,000 million liters to 9,600 million liters after installing efficient irrigation systems.

What is the percent decrease in annual water use?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI18-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a protein-rich breakfast affects hunger before lunch. Group A eats a protein-rich breakfast. Group B eats a lower-protein breakfast with the same number of calories. Hunger is rated three hours later using the same survey.

Why should both breakfasts have the same number of calories?

- A) To help isolate protein amount as the main variable being tested
- B) To make protein and calories identical substances
- C) To guarantee every person reports the same hunger level
- D) To prevent the researcher from collecting survey data

GED-SCI18-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the texture of a shoe sole affects traction. The same shoe model is made with four sole textures. Each shoe is pulled across the same test surface, and the force needed to make the shoe begin sliding is measured.

What is the dependent variable?

- A) Force needed to make the shoe begin sliding
- B) Texture of the shoe sole
- C) Same test surface
- D) Same shoe model

GED-SCI18-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A wedge is a simple machine that changes an input force into forces that push material apart. A knife blade is an example of a [D1]. A sharper blade can require [D2] input force to cut the same material.

Dropdowns:

D1) wedge | ribosome | weather front

D2) less | inherited | more orbital

GED-SCI18-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A physical therapist records average knee range of motion after two rehabilitation plans.

Plan | Average knee range of motion after 8 weeks

Standard therapy | 112°

Standard therapy plus strength training | 126°

Which claim is best supported by the data?

- A) Adding strength training was associated with greater knee range of motion.
- B) Standard therapy alone produced greater knee range of motion.
- C) Both plans produced exactly the same range of motion.
- D) Range of motion cannot be measured in degrees.

GED-SCI18-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the force needed to cut the same material with blades of different edge angles.

Blade edge angle | Force needed to cut (N)

20° | 18

30° | 25
40° | 34
50° | 46

Which statement is best supported by the data?

- A) Smaller edge angles required less force to cut the material.
- B) Larger edge angles required less force to cut the material.
- C) Edge angle had no relationship to cutting force.
- D) Every blade required exactly 46 N.

GED-SCI18-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A county tracks reclaimed mine land and stream acidity near former mining sites.

Year | Reclaimed mine land (hectares) | Average stream pH

1 | 120 | 5.2

2 | 190 | 5.8

3 | 260 | 6.3

4 | 340 | 6.8

Which TWO statements are best supported by the data?

- A) Reclaimed mine land increased from Year 1 to Year 4.
- B) Average stream pH moved closer to neutral from Year 1 to Year 4.
- C) Reclaimed mine land decreased every year.
- D) Average stream pH became more acidic every year.
- E) Both measurements stayed exactly the same.

GED-SCI18-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person has a change in a gene that affects a protein channel in kidney cells. The channel normally helps reabsorb water and dissolved substances. If the channel works poorly, the kidneys may have difficulty maintaining proper fluid balance.

Which statement best explains the effect?

- A) A gene change can alter a kidney cell protein and affect fluid balance.
- B) Kidney cells do not use proteins.
- C) Genes affect only minerals in rocks.
- D) Fluid balance is unrelated to cell membranes.

GED-SCI18-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A metal safety barrier bends during a crash. The bending increases the time over which the vehicle slows down and spreads energy transfer over a longer time.

Why can this design reduce injury risk?

- A) Increasing stopping time can reduce the average force on the vehicle.
- B) Increasing stopping time removes all momentum instantly.
- C) Bending metal prevents energy transfer.
- D) Bending metal stops gravity from acting.

GED-SCI18-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist studies a region where a river once carried gravel into a basin. Over time, the gravel was buried, compacted, and cemented into a rock made of rounded pebbles.

Which rock is most likely described?

- A) Conglomerate
- B) Basalt
- C) Marble
- D) Obsidian

GED-SCI18-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A medical study includes 720 adults. Of these adults, 126 report a family history of kidney disease.

What percent of the adults report a family history of kidney disease?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI18-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how handle length affects the force needed to turn a screwdriver. The same screw and screwdriver tip are used in every trial. Only the handle length changes, and the turning force is measured.

Which factor is the independent variable?

- A) Handle length

- B) Turning force
- C) Same screw
- D) Same screwdriver tip

GED-SCI18-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The smooth endoplasmic reticulum helps make lipids and helps some liver cells process certain drugs and toxins. These functions support cell membranes and detoxification.

Which statement best describes the smooth endoplasmic reticulum?

- A) It helps make lipids and process certain toxins.
- B) It controls plate motion at mid-ocean ridges.
- C) It reflects sunlight from a planet's surface.
- D) It stores all urine before it leaves the body.

GED-SCI18-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the output speed of a gear system when the input speed is changed.

Input speed (rotations/min): 20, 40, 60, 80

Output speed (rotations/min): 10, 20, 30, 40

Which statement is best supported by the data?

- A) Output speed was half of input speed in each trial.
- B) Output speed was twice the input speed in each trial.
- C) Output speed stayed at 10 rotations/min in every trial.
- D) Input speed had no relationship to output speed.

GED-SCI18-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four seismic waves recorded from the same earthquake.

Wave | Arrival time after earthquake starts (min)

- A | 4
- B | 7
- C | 12
- D | 19

Which wave arrived first?

- A) Wave A
- B) Wave B
- C) Wave C
- D) Wave D

GED-SCI18-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records average blood creatinine, a waste product used to evaluate kidney function, before and after a treatment.

Time | Average blood creatinine (mg/dL)

Before treatment | 2.4

After 4 weeks | 1.8

After 8 weeks | 1.3

Which statement is best supported by the data?

- A) Average blood creatinine decreased after treatment.
- B) Average blood creatinine increased after treatment.
- C) Creatinine stayed exactly the same.
- D) The treatment made blood contain no waste products.

GED-SCI18-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places two tuning forks near each other. When one tuning fork vibrates at the same natural frequency as the other, the second tuning fork begins vibrating too.

Which wave behavior is mainly described?

- A) Resonance
- B) Condensation
- C) Chromosome mutation
- D) Rock cementation

GED-SCI18-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether different desk heights affect neck muscle strain during computer work. Participants use low, medium, and high desk settings. Neck muscle activity is measured with the same sensor during each condition.

Which TWO steps would improve the investigation?

- A) Use the same muscle-activity sensor for all conditions.
- B) Keep the computer task the same for each desk height.
- C) Change both desk height and screen brightness together.
- D) Record only the lowest muscle-strain value.
- E) Use a different measurement method for each condition.

GED-SCI18-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A wave travels at 48 m/s and has a frequency of 12 Hz.

Wave speed = frequency $\times$ wavelength

What is the wavelength?

- A) 4 m
- B) 12 m
- C) 36 m
- D) 576 m

GED-SCI18-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A field is abandoned after farming stops. Soil remains, along with some seeds and roots. In the first years, grasses and wildflowers grow. Later, shrubs and young trees appear as the community changes.

Which ecological process is described?

- A) Secondary succession
- B) Lunar eclipse
- C) Acid-base neutralization
- D) Hydraulic force transfer

GED-SCI18-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some cells have receptors that bind specific chemical signals. When the correct signal binds, the receptor changes shape and starts a response inside the cell. If the receptor is missing, the cell may not respond to the signal.

Which statement best describes receptors?

- A) They help cells detect and respond to specific chemical signals.

- B) They form folded mountains at plate boundaries.
- C) They turn sound waves into minerals.
- D) They prevent all proteins from functioning.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI18-001 — Correct: A — Correct Answer: Sleeping with a small lamp on was associated with lower sleep quality. — Explanation: Group B slept with a lamp on and had a lower average sleep quality score than Group A.

GED-SCI18-002 — Correct: C — Correct Answer: Object C — Explanation: Potential energy = weight $\times$ height. Object C has $12 \text{ N} \times 9 \text{ m} = 108 \text{ J}$, the greatest value listed.

GED-SCI18-003 — Correct: A — Correct Answer: Stream pH moves closer to neutral and vegetation increases near streambanks. — Explanation: The cleanup aims to reduce acidic runoff and restore vegetation.

GED-SCI18-004 — Correct: A — Correct Answer: It helps coordinate homeostatic responses. — Explanation: The hypothalamus helps regulate internal conditions such as temperature, hunger, thirst, and hormone signaling.

GED-SCI18-005 — Correct: 140 — Tolerance: exact — Correct Answer: 140 — Explanation: Gravitational potential energy = $35 \text{ N} \times 4 \text{ m} = 140 \text{ J}$.

GED-SCI18-006 — Correct: A,B — Correct Answer: Car body | Surface used for testing — Explanation: These factors should stay constant so wheel diameter is the tested variable.

GED-SCI18-007 — Correct: D1=karyotype; D2=chromosome abnormalities — Correct Answer: D1=karyotype; D2=chromosome abnormalities — Explanation: A karyotype displays chromosomes and can reveal abnormalities in chromosome number or structure.

GED-SCI18-008 — Correct: A — Correct Answer: Light travels in straight paths unless it is reflected, absorbed, or bent. — Explanation: A shadow forms because an opaque object blocks light traveling from the source.

GED-SCI18-009 — Correct: A — Correct Answer: Earth's outer core is liquid. — Explanation: S waves do not travel through liquids, so their absence through the outer core supports that it is liquid.

GED-SCI18-010 — Correct: A — Correct Answer: High-intensity cycling was associated with higher blood lactate. — Explanation: The high-intensity session had a higher average lactate level, 6.4 mmol/L.

GED-SCI18-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Length of the crank handle; T2=Force needed to turn the crank; T3=Same load attached to the crank system — Explanation: Crank length is changed, turning force is measured, and the attached load is controlled.

GED-SCI18-012 — Correct: A — Correct Answer: Sensing body position and helping maintain balance — Explanation: Inner-ear structures send balance and position information to the brain.

GED-SCI18-013 — Correct: A — Correct Answer: Elastic potential energy changing into kinetic energy — Explanation: The stretched rubber band stores elastic potential energy that becomes kinetic energy when released.

GED-SCI18-014 — Correct: 20 — Tolerance: exact — Correct Answer: 20 — Explanation: Water use decreased by 2,400 million liters. $2,400 \div 12,000 \times 100 = 20\%$.

GED-SCI18-015 — Correct: A — Correct Answer: To help isolate protein amount as the main variable being tested — Explanation: Equal calories help test whether protein level, not calorie amount, affects hunger.

GED-SCI18-016 — Correct: A — Correct Answer: Force needed to make the shoe begin sliding — Explanation: The dependent variable is the measured force after changing sole texture.

GED-SCI18-017 — Correct: D1=wedge; D2=less — Correct Answer: D1=wedge; D2=less — Explanation: A knife is a wedge, and a sharper wedge can require less input force to cut.

GED-SCI18-018 — Correct: A — Correct Answer: Adding strength training was associated with greater knee range of motion. — Explanation: The combined plan had a greater average range of motion, 126° , than standard therapy, 112° .

GED-SCI18-019 — Correct: A — Correct Answer: Smaller edge angles required less force to cut the material. — Explanation: The force needed increased as blade edge angle increased.

GED-SCI18-020 — Correct: A,B — Correct Answer: Reclaimed mine land increased from Year 1 to Year 4. | Average stream pH moved closer to neutral from Year 1 to Year 4. — Explanation: Reclaimed land increased from 120 to 340 hectares, and pH rose from acidic 5.2 toward neutral 7.

GED-SCI18-021 — Correct: A — Correct Answer: A gene change can alter a kidney cell protein and affect fluid balance. — Explanation: A mutation can change a channel protein, affecting kidney cell transport and fluid balance.

GED-SCI18-022 — Correct: A — Correct Answer: Increasing stopping time can reduce the average force on the vehicle. — Explanation: Spreading the stopping process over more time can reduce average force.

GED-SCI18-023 — Correct: A — Correct Answer: Conglomerate — Explanation: Conglomerate is a sedimentary rock made of rounded pebbles cemented together.

GED-SCI18-024 — Correct: 18 — Tolerance: exact — Correct Answer: 18 — Explanation: $126 \div 720 \times 100 = 17.5\%$, which rounds to 18%.

GED-SCI18-025 — Correct: A — Correct Answer: Handle length — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI18-026 — Correct: A — Correct Answer: It helps make lipids and process certain toxins. — Explanation: The smooth endoplasmic reticulum is involved in lipid production and detoxification processes.

GED-SCI18-027 — Correct: A — Correct Answer: Output speed was half of input speed in each trial. — Explanation: The output speed is consistently one-half of the input speed.

GED-SCI18-028 — Correct: A — Correct Answer: Wave A — Explanation: Wave A arrived first because it has the shortest arrival time, 4 minutes.

GED-SCI18-029 — Correct: A — Correct Answer: Average blood creatinine decreased after treatment. — Explanation: Average creatinine decreased from 2.4 to 1.3 mg/dL after treatment.

GED-SCI18-030 — Correct: A — Correct Answer: Resonance — Explanation: Resonance occurs when one vibrating object causes another object with the same natural frequency to vibrate.

GED-SCI18-031 — Correct: A,B — Correct Answer: Use the same muscle-activity sensor for all conditions. | Keep the computer task the same for each desk height. — Explanation: Consistent measurement and task conditions improve the investigation.

GED-SCI18-032 — Correct: A — Correct Answer: 4 m — Explanation: Wavelength = wave speed $\div$ frequency = $48 \text{ m/s} \div 12 \text{ Hz} = 4 \text{ m}$.

GED-SCI18-033 — Correct: A — Correct Answer: Secondary succession — Explanation: Secondary succession occurs when a community regrows where soil, seeds, or roots remain after disturbance.

GED-SCI18-034 — Correct: A — Correct Answer: They help cells detect and respond to specific chemical signals. — Explanation: Receptors bind specific signals and trigger cell responses.